IndiaKart E-Commerce Analytics

Phase 1 — Data Quality Report

This report summarizes the initial profiling and validation of the eight supplied CSV tables. The checks covered record counts, missing values, duplicates, selected date and numeric fields, relationships between orders and related tables, payment outcomes, inventory status, and high-value orders.

Dataset profile and key checks

Table	Rows	Key findings
customers	10,000	No missing values; no duplicate rows or customer IDs. 457 customers (4.57%) have total_orders = 0.
orders	50,000	No missing values except delivered_date: 17,501 (35%). No duplicate rows or order IDs. 312 orders exceed ■5 lakh.
order_items	100,000	No missing values or duplicate rows. All order IDs match orders. No zero/negative quantities or unit prices.
products	1,000	No missing values or duplicate rows.
payments	50,000	refund_date is missing in all 50,000 rows (100%). 48,252 Success and 1,748 Failed; failure rate 3.50%. payment_date converted with 0 invalid/missing values.
returns	10,000	No missing values or duplicate rows. All return order IDs match orders.
inventory	1,000	No missing values or duplicate rows. Status counts: 922 In Stock, 76 Low Stock, 2 Out of Stock.
suppliers	200	No missing values or duplicate rows.

Issues and interpretation

Payment failures: The 3.50% failure rate is above the PRD's stated target of less than 2%. Investigate payment gateway, method, and failure reasons.

Missing refund dates: All payment refund_date values are missing, and refund_amount was previously observed as zero throughout. Confirm whether this is expected in the dataset and how refund information should be represented before imputing or changing values.

Missing delivery dates: The 17,501 missing delivered_date values were associated with Cancelled, Shipped, Returned, and Processing statuses. Delivered orders had no missing delivery dates and no delivery date earlier than order_date in the check performed. Confirm business rules for Returned orders.

High-value orders: 312 orders have final_amount > ■5,00,000. Treat these as review flags, not confirmed errors; validate line items, quantities, discounts, and GST before taking action.

Inventory and customers: Two inventory records are Out of Stock. The 457 customers with zero orders are not automatically data-quality errors and should be retained unless business rules indicate otherwise.

Recommended actions

• Investigate the 100% missing refund_date field and confirm refund data conventions before attempting to fill it.

• Review the 312 high-value orders against order-item totals and applicable tax/discount rules.

• Investigate the 3.50% payment failure rate, which exceeds the PRD target of under 2%.

• Review the two Out of Stock inventory records and determine whether replenishment is required.

Scope note: This report reflects checks run and outputs shared so far. High-value orders and missing fields are review flags, not confirmed errors. Full date-type validation across all tables and order-to-line-item amount reconciliation remain to be completed before claiming exhaustive validation.